

Swetha Vaidyanathan Sriram

✉ vsswetha@gmail.com ☎ 925-750-0169 🔗 [linkedin.com/in/swetha-sriram-a75098233/](https://www.linkedin.com/in/swetha-sriram-a75098233/)

🐙 github.com/swetha7502 🔗 swetha7502.github.io

EDUCATION

The Pennsylvania State University, Pennsylvania Master of Science, Computer Science	2025 – 2027 CGPA: 4/4
Madras Institute of Technology, Anna University, India Bachelor of Engineering, Computer Science and Engineering	2020 – 2024 CGPA: 3.94/4
Indian Institute of Technology, Madras, India Bachelor of Science, Data Science & Programming (Hybrid)	2020 – 2024 CGPA: 3.73/4

SKILLS

Languages & Database – Python, Java, C++/C, JavaScript, C#, Bash, Oracle SQL, MySQL, PostgreSQL, MongoDB, Redis, SQLite

Web Technologies – Flask, ReactJS, Vue.js, Next.js, TypeScript, Node.js, HTML5, CSS3, Bootstrap, REST APIs, JSP, FastAPI

AI & ML – LangSmith, RAG, LangChain, LangGraph, Scikit-Learn, TensorFlow, Keras, Tableau, OpenCV, PyTorch

Tools & Platforms – Oracle APEX, Git, ChromaDB, Jenkins, Docker, Kubernetes, OCI, AWS

WORK EXPERIENCE

Research Assistant | Pennsylvania State University 10/2025 – Present
• Modeling and predicting the behavior of water molecules with proteins using molecular dynamics simulations and numerical analysis using **C#** and **MatLab**.

Associate Applications Developer | Oracle Pvt. Ltd. 08/2024 – 08/2025
• Enhanced an enterprise-scale release management platform (100+ users across APAC and EMEA) by **designing and deploying 2 new workflows** using **PL/SQL, JavaScript, and Oracle APEX**, reducing manual scheduling (~20% improvement).
• Designed and developed a **Jira** ticket dashboard (150+ DAUs), reducing weekly status meetings by ~20% and accelerating ticket resolution time by 5%.
• Contributed to **3 Oracle SDK releases**, implementing core features and bug fixes that improved developer experience and ensured on-schedule delivery of enterprise updates.
• Resolved key bugs and wrote **JUnit** tests implemented improvements that increased system reliability and reduced release-related delays by ~10%.
• Analyzed and optimized **Oracle Cloud Infrastructure (OCI)** logs to identify and eliminate redundant entries. Resolved a Fortify-detected cybersecurity vulnerability by debugging and validating secure code implementation.

Project Intern | Oracle Pvt. Ltd. 01/2024 – 07/2024
• **Automated** regression **testing flows** with Oracle Utilities Testing Accelerator (**OUTA**) and **Selenium**, boosting test coverage by ~30% and reducing manual QA effort by ~40%.
• Built and deployed an energy forecasting model adopted by 10+ data science team members, improving prediction accuracy by ~20% over previous model using **Oracle ADB, Python, Java and Selenium**.

ACADEMIC PROJECTS

Medical Research Multi-Agent System 🔗
Built a **LangGraph**-orchestrated multi-agent pipeline with three specialized agents (Retriever, Fact-Checker, Synthesizer) that search **PubMed** and **arXiv** in real time, embed results into **ChromaDB** for semantic retrieval, and synthesize evidence-graded answers with citations. Backend in **FastAPI**, frontend in **Next.js 15** with **TypeScript**.

ConflictMind - AI Agent that Remembers You 🔗
Built an AI agent with persistent memory and adversarial conflict reconciliation using **Google ADK, Gemini 2.5 Flash via Vertex AI, Flask and MongoDB Atlas**; engineered a 3-step debate pipeline that detects contradictory user memories via vector similarity search + **LLM** classification, resolves them through structured Gemini debates, and writes unified memories back to Atlas, deployed on **Google Cloud Run** via **Docker** with **MongoDB MCP** server.

PixEdit : Text-Based Image Modification for Content Creation 🔗
Built a text-guided image editing system using **Grounded-DINO, SAM, and LaMa** inpainting; implemented object segmentation, removal, and recoloring pipelines with a **Flask REST API** backend running fully on CPU.